

UDIT KULKARNI

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Professional Summary

Software Engineer and AI Systems Architect with a Master's in Computer Applications (Machine Learning), specializing in **production AI, backend architecture, enterprise automation, and performance optimization**. Hands-on across Laravel/PHP and Python/FastAPI, with production experience in RAG, conversational AI, Text-to-SQL, CRM/ERP platforms, high-concurrency APIs, and enterprise integrations.

Core Skills

Languages: PHP, Python, C++, JavaScript

Backend: Laravel, Symfony, Drupal, FastAPI, Flask, REST APIs, WebSockets, SQL

AI / GenAI: Azure OpenAI, OpenAI, Azure AI Foundry, LangGraph, LangChain, RAG, Qdrant, FastEmbed, BM25, Hybrid Search, Cross-Encoder Reranking, BAAI/bge-reranker-base, Text-to-SQL, LangSmith

Testing & Frontend: PyTest, PHPUnit, HTML/CSS, Tailwind CSS, Figma-to-HTML workflows

Data / Cloud: MySQL, SQLite, Redis, AWS S3, AWS EC2, AWS Lambda, AWS ECS, AWS ECR, AWS CloudFront, Docker, Linux, Nginx, GitHub Actions, GitLab CI, Jenkins, CI/CD

Enterprise / Security: SAP APIs, Microsoft Graph API, OpenID/OAuth, JWT, RBAC, HITL Guardrails, BillDesk JWS (HMAC-SHA256), Server-Sent Events (SSE)

Architecture: System Design, Modular Monoliths, Microservices, State Machines, Asynchronous Queues, Multi-Tenant Systems, OOP, SOLID, Observability, Performance Optimization, Data Structures & Algorithms

Professional Experience

Software Developer

Publicis Digital Experience

HINCOL — Enterprise AI, CRM & Sales Platform

Jan 2025 – Present

HINCOL & Hinduja Hospital

B2B Manufacturing

- Built a **multi-criteria weighted Lead Scoring & Qualification engine** in Laravel using project type, volume, product specifications, margin and urgency, processing **800–1,500 enterprise leads/opportunities per month** across 5 national zones.
- Reduced lead qualification from **~15 minutes to <1 second** and routing from **24–48 hours to real time**, saving an estimated **60–80 hours/month** of sales and leadership effort and reducing administrative vetting overhead by **~35%**.
- Architected a **Key Account Management (KAM) classification engine** for **~2,000–2,500 active accounts**, replacing quarterly spreadsheet reviews with event-driven and queued batch re-evaluation, plus a multi-stage approval state machine with AWS S3 audit attachments.
- Engineered a **Product Penetration & Cross-Sell matrix engine** to calculate customer whitespace deficits from sales orders, enabling sales teams to identify product-category gaps across customer accounts.
- Integrated **Microsoft Graph API** for automated Outlook seminar calendar synchronization.
- Contributed to a **77+ module Laravel enterprise platform** spanning leads, opportunities, RFQs, indents and credit governance, with SAP integrations for pricing, inventory, sales orders, bank guarantees and statements.

Hincol – Voice Assistant AI

- Architected and built **HincolBot**, an enterprise full-duplex conversational voice platform using FastAPI, WebSockets and Azure AI Foundry, reducing ERP/CRM data-entry time by **80%** across 5 national operating zones, from 10–15 minutes to **<2 minutes** per opportunity.
- Engineered bidirectional **24kHz PCM16 speech streaming** over authenticated WebSockets using GPT-Live-1 and browser Web Audio APIs, achieving **<200ms audio response latency** with real-time barge-in, interruption handling, Redis conversational memory and persistent session summaries.

Strategic Sales Agent

- Architected a deterministic **LangGraph StateGraph** with checkpointed multi-turn context, intent classification and dynamic slot filling, physically separating conversational reasoning from database execution and pausing for confirmation before mutations.
- Engineered **hybrid semantic entity disambiguation** across **4,628 master vectors** spanning ~2,500 contractors/customers, ~1,500 locations, 20 product grades and 619 highway projects; fused Qdrant dense retrieval with lexical SQL search to achieve **98.4% accuracy across 500 enterprise conversational test cases**.

- Developed governed **Text-to-SQL** with vector schema discovery, read-only MySQL validation, sensitive-column masking and SSE streaming; added asynchronous Redis workflows and **LangSmith** observability for p50/p90/p99 latency, token usage, tracing and continuous AI evaluation.
- Implemented defense-in-depth **HITL security**: multilingual Zero-Delete protection, prompt-injection defenses, PII/schema validation and zone-scoped RBAC across 5 operating regions, requiring explicit confirmation before ACID database mutations.

Hinduja Hospital — Healthcare RAG & Conversational AI

Healthcare / HIS

- Architected an enterprise **Healthcare RAG & Conversational AI Agent** using Python 3.11, FastAPI, Qdrant and LangChain for P.D. Hinduja Hospital (Mahim & Khar), integrating **260+ doctors, 90 specialties and 2 hospital units** for symptom triage, doctor discovery and real-time OPD booking.
- Engineered a two-stage **Hybrid Search pipeline** over **3,096 clinical chunks**, combining dense retrieval and BM25 to maximize recall across symptom language and exact medical entities, followed by **BAAI/bge-reranker-base** cross-encoder reranking for precision and clinical grounding.
- Designed query-adaptive retrieval weighting (70/30 dense-sparse) for symptom versus doctor/department queries, reducing a **~30-candidate pool to 6–12 grounded chunks** through reranking, threshold gating and token-budget allocation.
- Built an asynchronous **multi-turn state machine and HIS integration layer** supporting live slot discovery, SMS OTP authentication, BillDesk payment processing using JWS HMAC-SHA256 and automated hospital voucher generation; validated same-slot contention with **250 simultaneous booking attempts**.
- Validated the platform through a **320+ automated PyTest/PHPUnit suite**, achieving **96.6% pass rate for Mahim and 86.9% for Khar** in automated clinical and functional regression tests; end-to-end booking flows completed in **~45–60 seconds** in interactive testing.

Software Developer & Delivery

Jan 2023 – Dec 2024

Razorfish

Disney+ Hotstar & Citroën

- Architected and delivered the **Citroën Dealer Locator REST API**, aggregating real-time Stellantis/PSA global APIs, Citroën Advisor reviews and localized databases; supported a platform serving **1.5M+ monthly visitors** with **50+ RPS peak traffic** during national campaigns.
- Engineered low-latency geo-spatial dealer matching with real-time Haversine distance calculations, achieving **<350ms typical response latency**; deployed scalable containerized architecture using **Docker, AWS ECS/ECR, S3, CloudFront and Redis**.
- Developed **12+ REST API modules** for the Citroën India digital platform using Drupal and PHP, including automated vehicle-data ingestion and synchronization across **1,000+ dealer locations**.
- Engineered a custom **Disney+ Hotstar campaign CMS** for drafting emailers and an in-house project-management tool for delivery tracking, invoicing and pending-payment management.
- Integrated OTP verification, WhatsApp messaging and OpenID/OAuth authentication with third-party platforms, and improved page-load performance by **25%** through server-side caching and media-asset optimization.

Early Career

Software Developer Intern

Jan 2022 – Jul 2022

Publicis Media

View Certificate

- Contributed to PHP e-commerce backend development, modular architecture and SQL optimization for high-concurrency operations.

Education

Master's in Computer Applications (Machine Learning)

2021 – 2022

Sardar Patel Institute of Technology (SPIT), Mumbai

Bachelor of Science in Computer Science

2017 – 2020

DG Ruparel College of Arts, Science and Commerce, Mumbai

Certifications & Recognition

- **Wow@Work Collective Impact Award** — HINCOL platform delivery *View Award*
- **Master RAG: Retrieval-Augmented Generation Systems** — Udemy, Sept 2026 *Verify*
- **Software Architecture & Design of Modern Large Scale Systems** — Udemy *Verify*
- **AI-900: Microsoft Azure AI Fundamentals** — Microsoft / Udemy *Verify*
- **GitHub Copilot for Developers: Code Faster With AI Agents** — Udemy *Verify*
- **Microsoft Technology Associate (MTA): Security Fundamentals** — Microsoft *View Certificate*